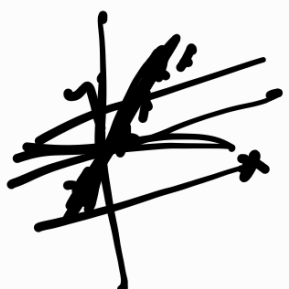
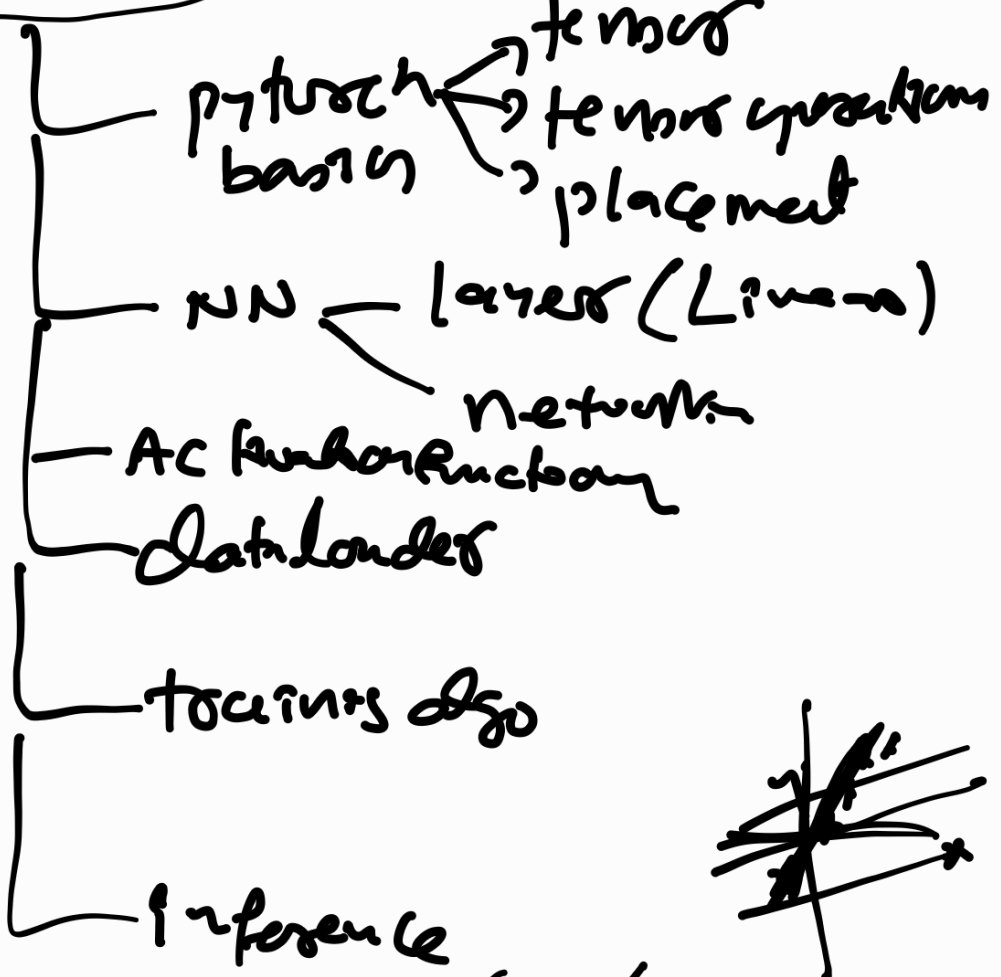
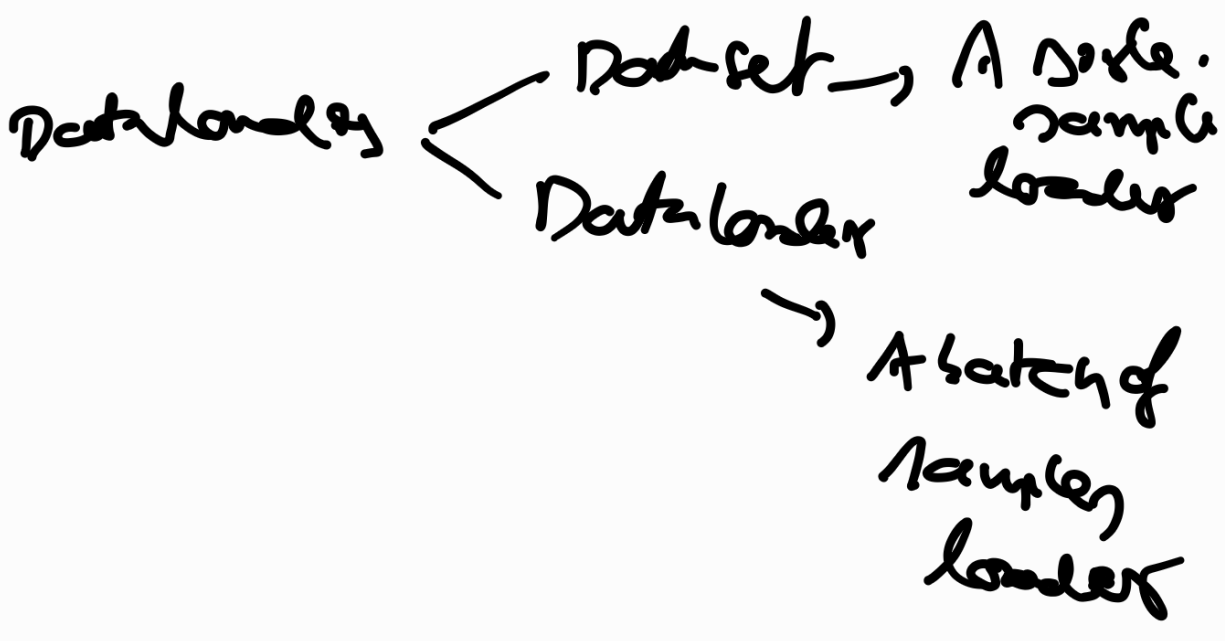
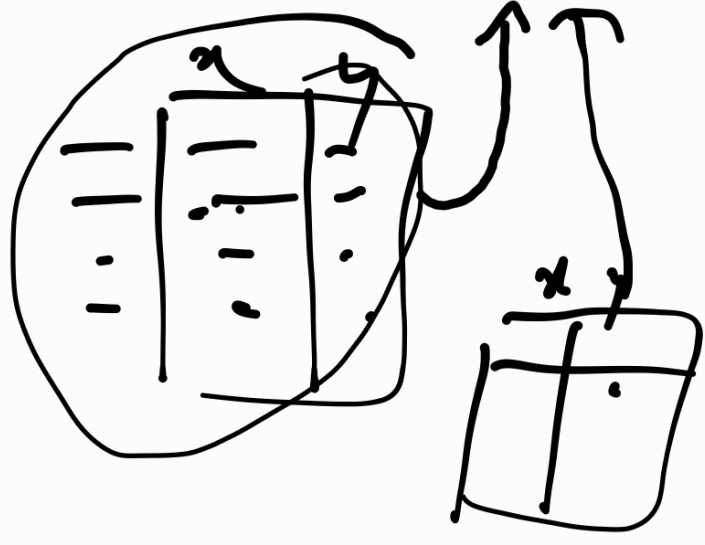
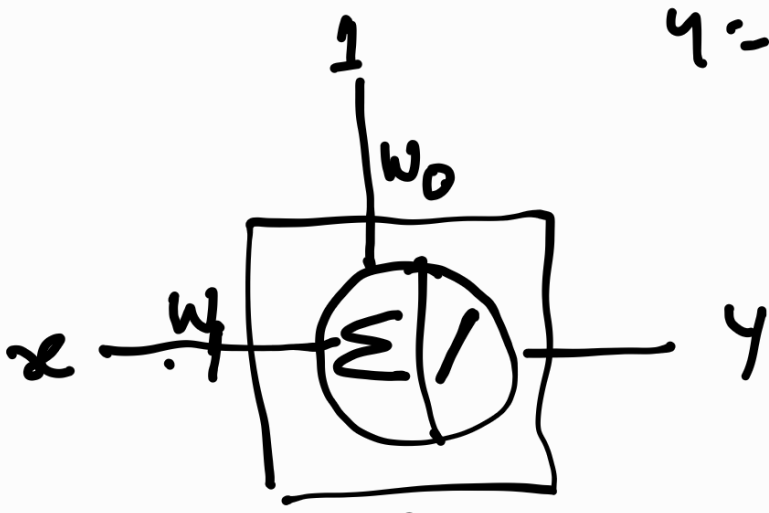


Basic NN

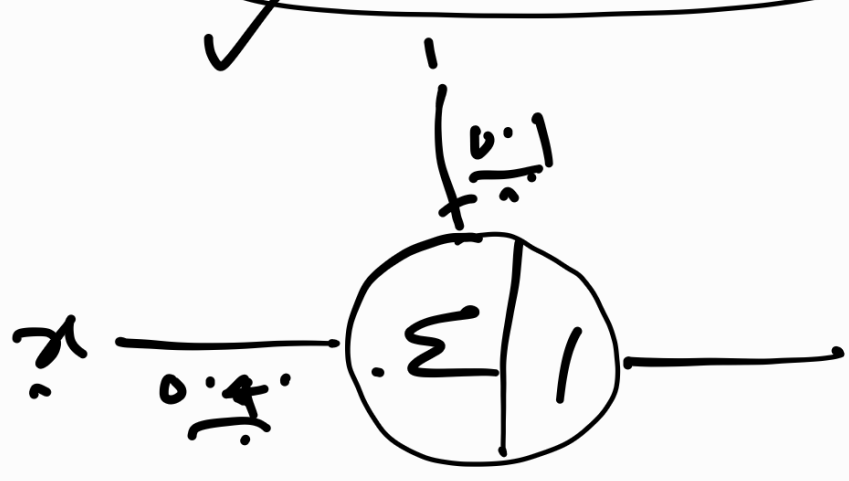
array/l31



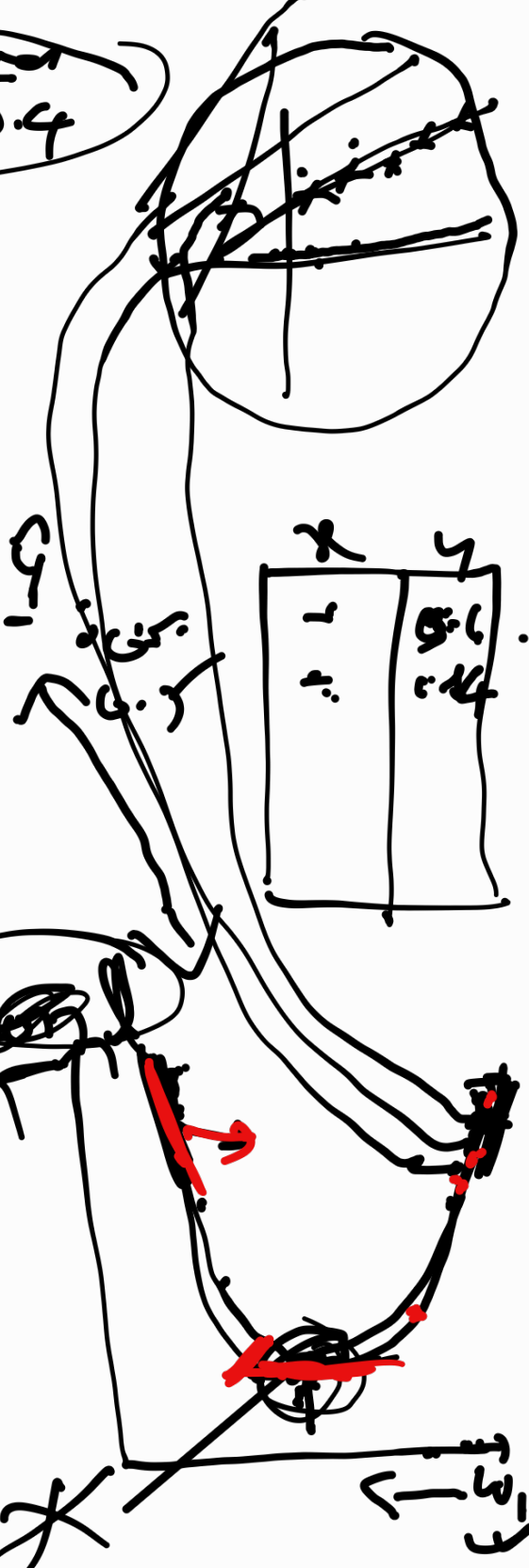
$$y = \textcircled{w_0} + \textcircled{w_1}x$$



$$w_0 = \frac{\text{random}}{0.1} \quad w_1 = \frac{\text{random}}{0.4}$$



x	y
1	0.6
2	0.4



or

0.1 -0.1 0.3

-0.3

1 0.5 1.0 1.5 2.0

mean/median/mode

mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

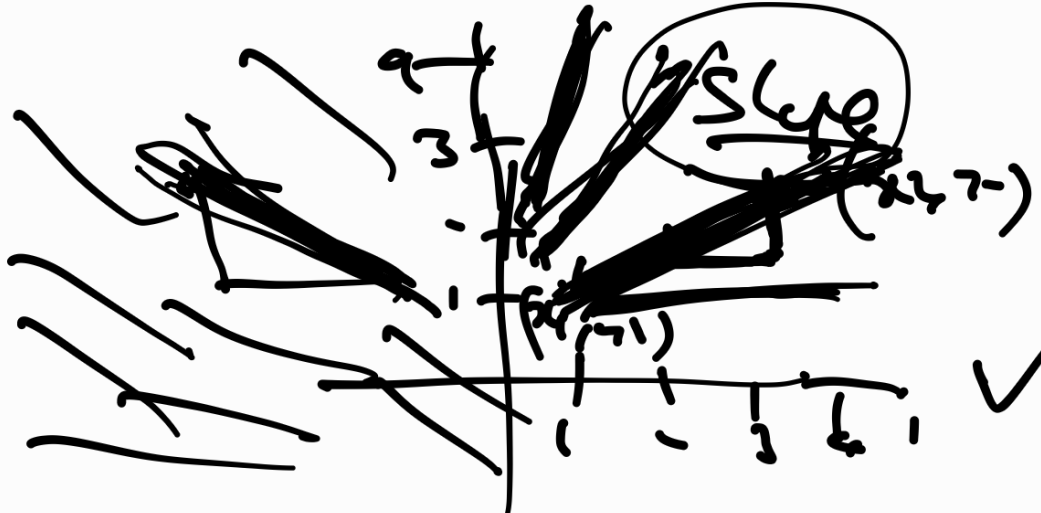
Mean Squared Error

$$J(w_0, w_1) = \frac{1}{2} \sum_{i=1}^n (y_i - w_0 - w_1 x_i)^2$$

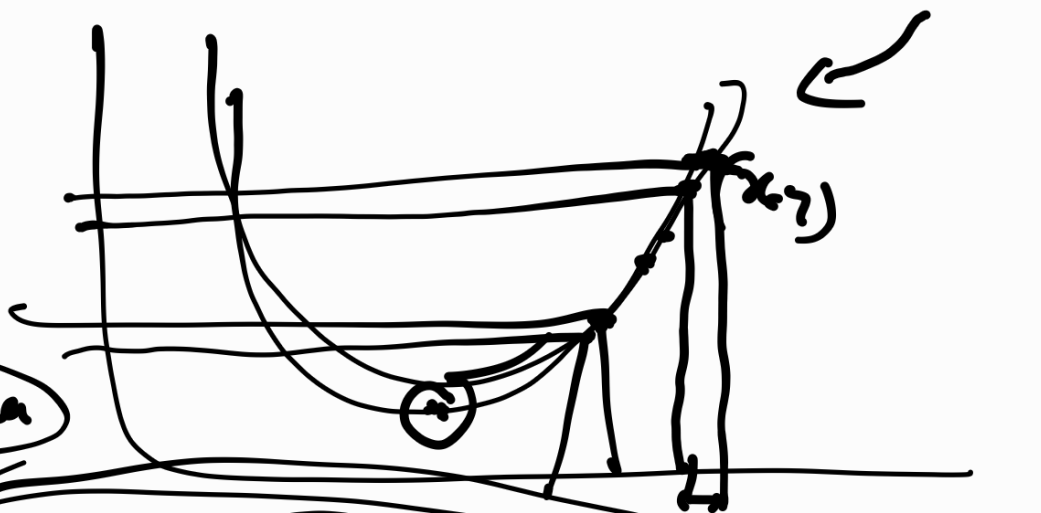
$w_0, w_1 =$

$$\hat{w}_1 = w_1 - \eta \frac{dJ}{dw_1}$$

neural network learning



$$x_2 - x_1$$



Approx

$$\lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x) - y}{(x + \Delta x) - x}$$

$\Delta x \rightarrow 0$
 $\Delta x : 0$

derivative

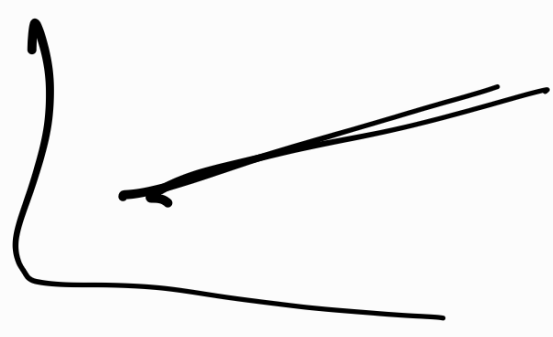
$$\frac{dy}{dx}$$

$$y = x^2 + 2$$

$$\frac{dy}{dx} = 2x$$

$$y = 2x + 2$$

$$\frac{dy}{dx} = 2$$

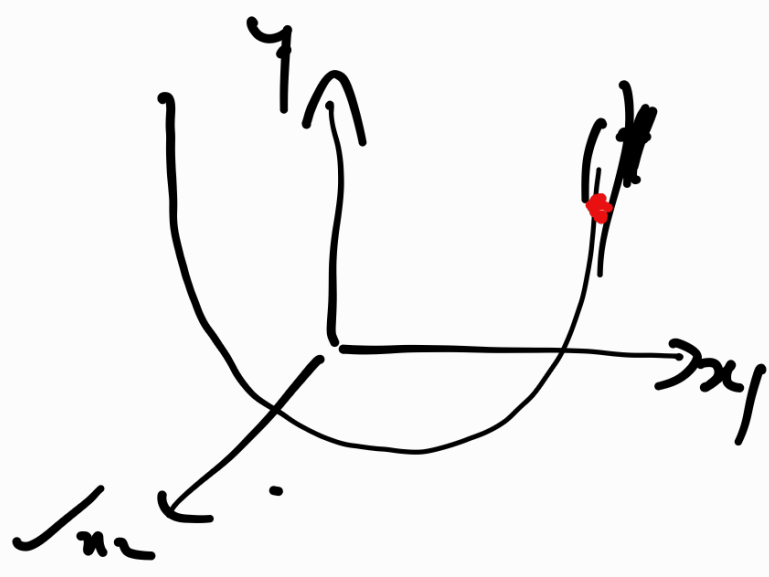


$$\frac{\partial y}{\partial x_1}$$

x_1 is const

$$\frac{\partial y}{\partial x_2}$$

x_2 is const



$$\frac{\partial \gamma}{\partial x_1}$$

x_1, x_2, x_3 can

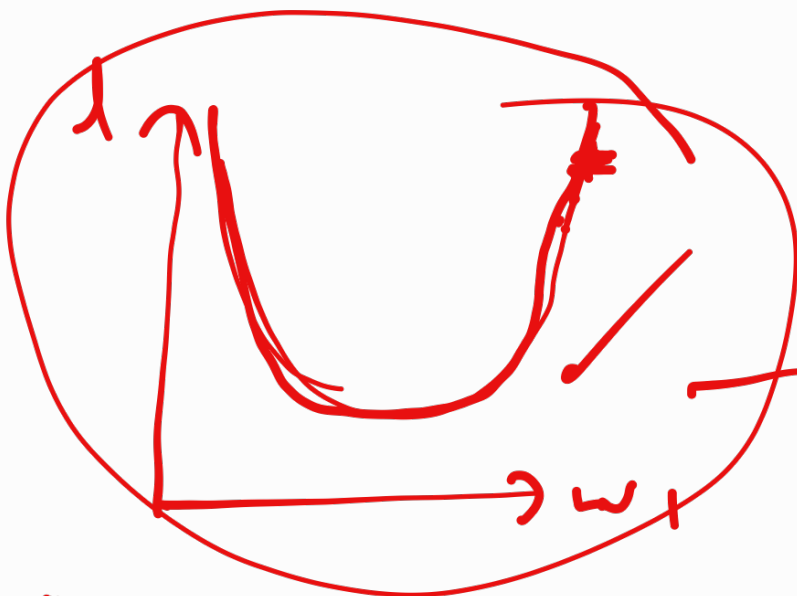


$$\frac{\partial \gamma}{\partial x}$$

x_1, x_2, x_3

gradient = vector of
partial derivatives

$$\nabla \gamma(x_1, x_2, x_3) = \left[\frac{\partial \gamma}{\partial x_1}, \frac{\partial \gamma}{\partial x_2}, \frac{\partial \gamma}{\partial x_3} \right]$$

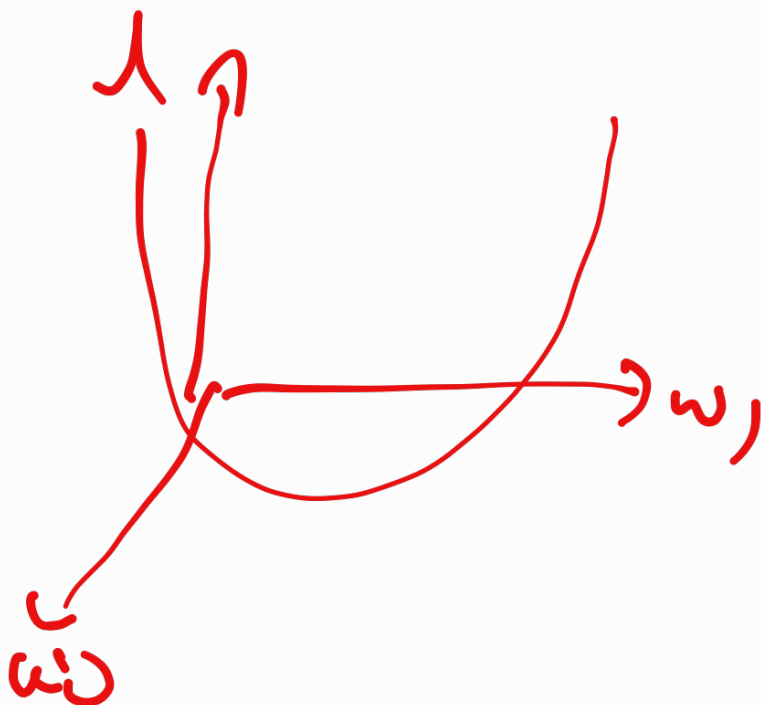


$$w_1 = -$$

$$w_1 = w_1 - \frac{\partial \gamma}{\partial w_1}$$

either slope = 0
slope = 0.0001

$$w_1 = w_1 - \eta \times \frac{\partial \gamma}{\partial w_1}$$



$$w_0 = \dots, w_1 = \dots$$

$$w_1 = w_1 - \eta \times \frac{\partial J}{\partial w_1}$$

$$w_0 = w_0 - \eta \times \frac{\partial J}{\partial w_0}$$

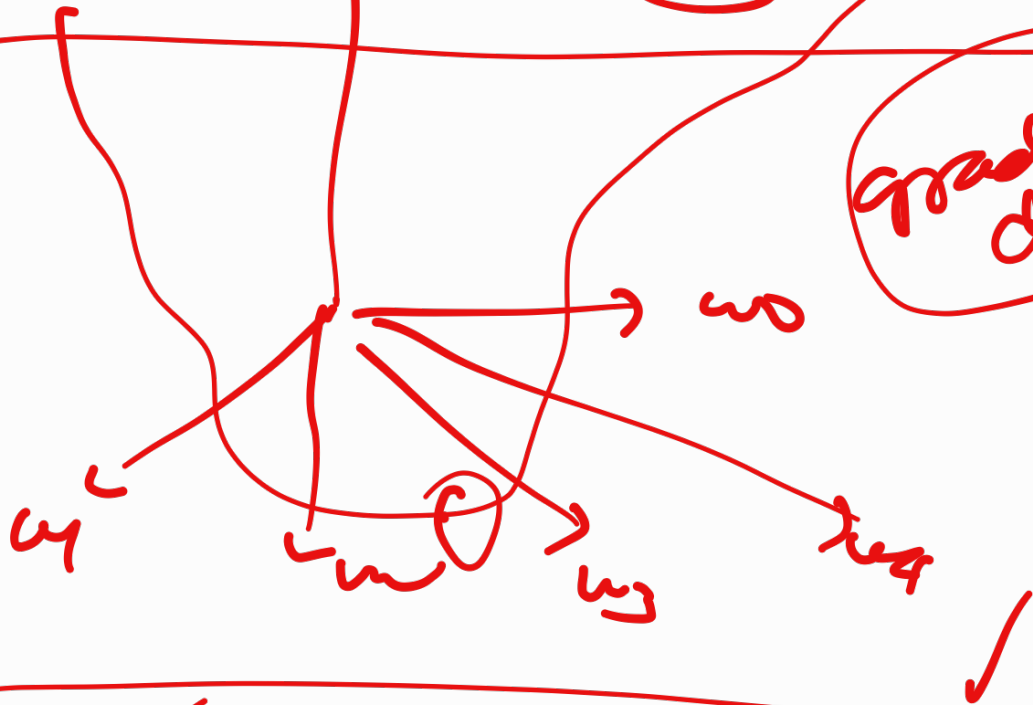
(50)

$$(w_0, w_1) = (-, -)$$

$$(w_0, w_1) = (w_0, w_1) - \eta \times \left(\frac{\partial J}{\partial w_0}, \frac{\partial J}{\partial w_1} \right)$$

$$(w_0, w_1) = (w_0, w_1) - \eta \times \left(\sum_{i=1}^n (y_i - \hat{y}_i) \right)$$

Gradient descent



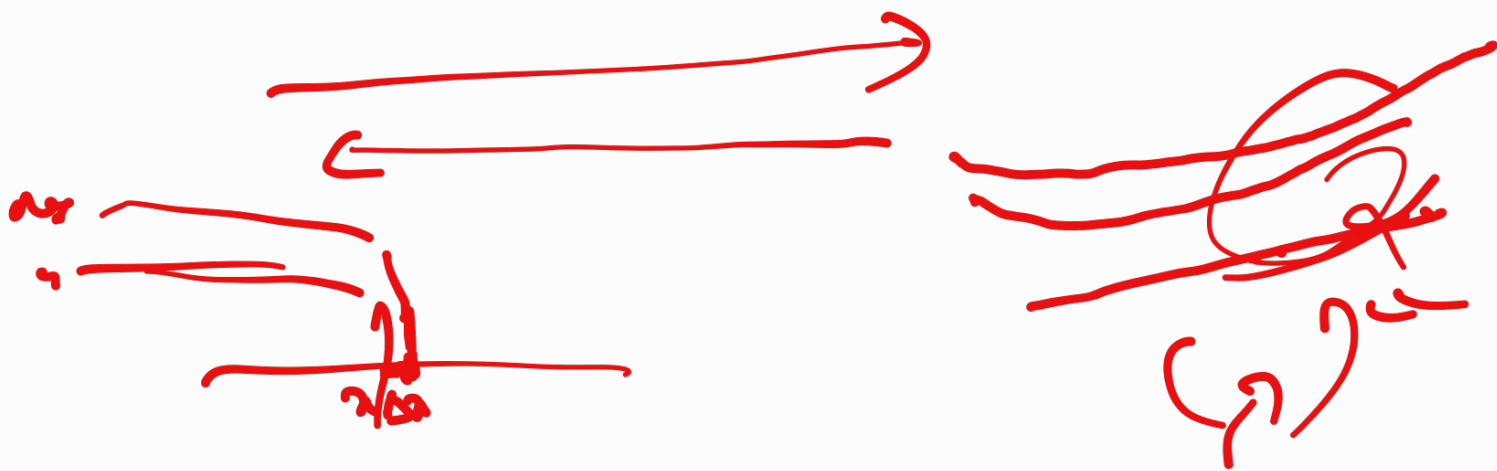
$$(w_0, w_1, w_2, \dots) = (-, -, -, \dots)$$

$$(w_0, w_1, w_2, \dots) = (w_0, w_1, \dots) - \eta \times \left(\frac{\partial J}{\partial w_0}, \frac{\partial J}{\partial w_1}, \dots \right)$$

key



DD DD
find pair
K
second pair



✓
Stochastik ~~so~~ descent
↖
batch size
↗
descent

$$l(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

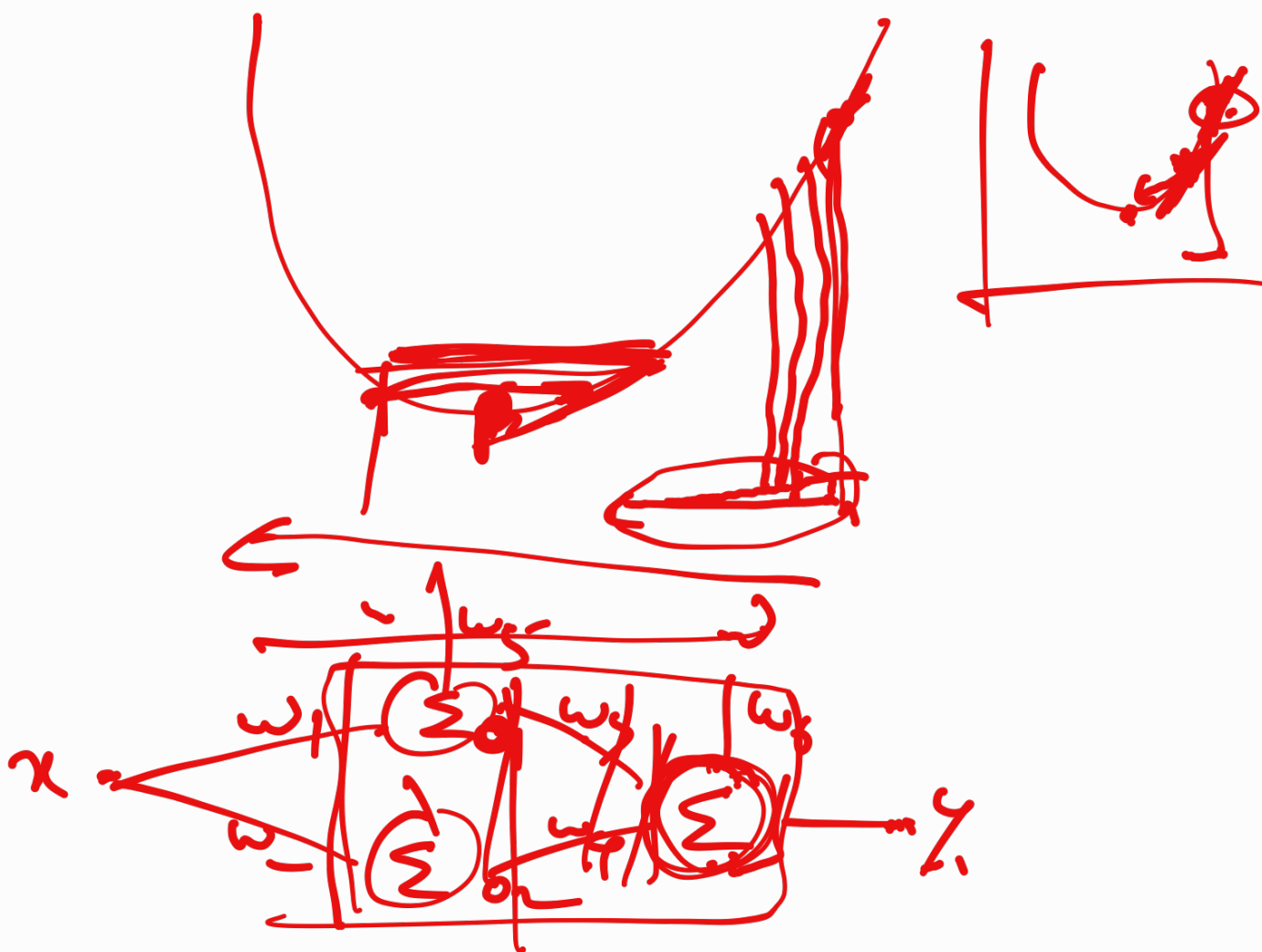
$$l(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - w_0 f(w_1, x_i))$$

$$(w_0, w_1) = (-, -)$$

$$\eta = 0.001$$

$$(w_0, w_1) = (w_0, w_1) - \eta \times \begin{pmatrix} \frac{\partial l}{\partial w_0} \\ \frac{\partial l}{\partial w_1} \end{pmatrix}$$

$$(w_0, w_1) = (w_0, w_1) - \eta \times \begin{pmatrix} \sum_{i=1}^n (y_i - w_0 - w_1 x_i) \\ \sum_{i=1}^n (y_i - w_0 - w_1 x_i) x_i \end{pmatrix}$$



linear layer

$$y = (w_1 w_3 + w_2 w_4) + w_6$$

~~$y = (w_1 x + w_2 w_3) + w_6$~~

permonats codes

- not to keep
- learn at specific step

< 153



$$y = \max_c$$



w_3



country

+

rank

State

sensor 4: 255

NN

Plan/Party (no E)

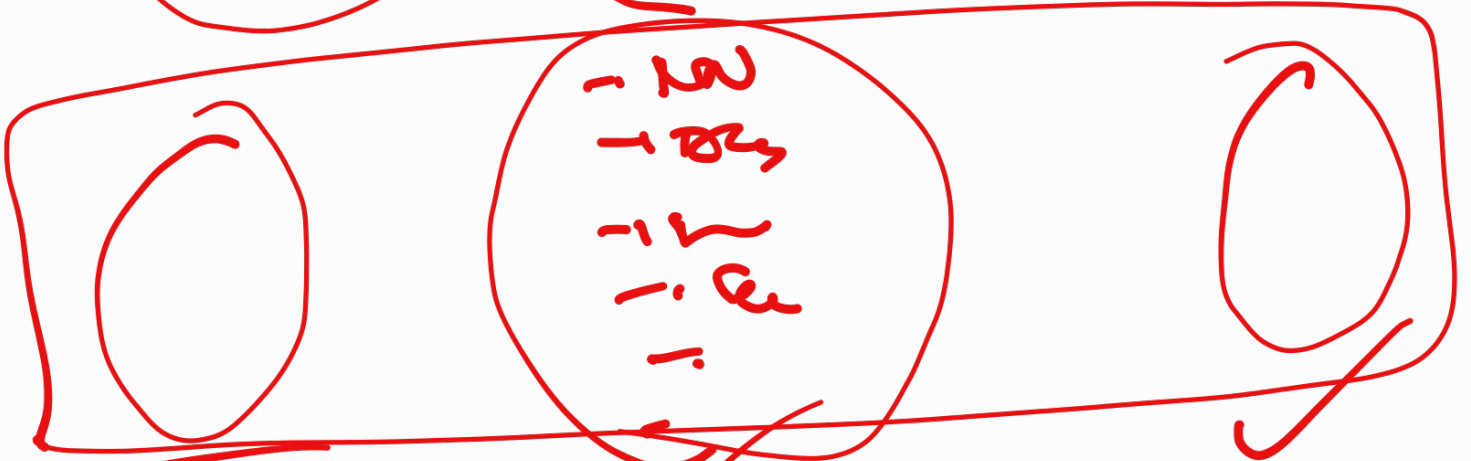
33b - 3b

pho/hy/hu/ae/H/cu

epa nate



Accuracy vs. Latency



Q

14